

CARNARVON, Australia

WMO: 943000

Lat: 24.8878S

Lon: 113.6700E

Elev: 5

StdP: 101.27

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.1	9.4	-6.6	2.2	22.3	-3.5	2.8	20.1	10.5	20.3	9.6	20.1	3.7	130	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.4	36.4	21.2	34.2	21.8	32.3	22.2	25.8	30.5	25.2	29.8	24.7	29.2	6.5	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.6	19.7	28.2	23.9	18.8	28.2	23.3	18.0	27.9	80.1	30.6	77.5	29.8	75.2	29.2	32.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.5	10.9	10.3	DB	5.5	41.8	1.1	2.5	4.7	43.6	4.1	45.1	3.5	46.5	2.7	48.3
			WB	2.7	27.2	1.2	1.7	1.8	28.4	1.1	29.4	0.5	30.4	-0.4	31.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	22.2	26.7	27.6	27.0	24.2	20.7	17.8	16.7	17.5	19.1	21.2	23.3	25.0	(6)
(7)		DBStd	4.31	2.31	2.36	2.43	2.30	2.25	2.03	1.96	1.82	1.93	1.85	2.09	2.18	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	149	0	0	0	0	4	33	59	38	13	1	0	0	(9)
(10)		CDD10.0	4457	517	493	526	426	333	235	207	233	273	348	400	466	(10)
(11)		CDD18.3	1563	259	259	267	176	79	18	8	13	36	91	150	208	(11)
(12)		CDH23.3	12587	2381	2682	2664	1283	475	101	49	86	149	386	831	1501	(12)
(13)	CDH26.7	4117	818	1039	1014	361	106	8	3	8	32	88	218	421	(13)	
(14)	Wind	WSAvg	6.1	7.5	7.0	6.5	5.6	4.6	4.4	5.0	6.2	7.1	7.4	7.8	(14)	
(15)	Precipitation	PrecAvg	213	9	21	16	15	31	46	47	17	6	4	3	1	(15)
(16)		PrecMax	402	69	148	210	89	143	160	215	58	28	38	48	14	(16)
(17)		PrecMin	75	0	0	0	0	0	1	0	0	0	0	0	0	(17)
(18)		PrecStd	97	16	33	33	21	36	34	53	15	7	7	7	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.3	39.2	38.9	36.0	32.1	27.8	27.0	27.7	30.3	32.9	36.0	36.8	(19)
(20)			MCWB	21.5	22.1	21.6	18.2	16.3	14.5	14.4	13.9	14.9	17.2	18.1	19.5	(20)
(21)		2%	DB	34.7	35.5	35.4	32.3	29.5	25.9	24.7	25.6	26.7	28.9	31.7	32.7	(21)
(22)			MCWB	23.2	23.4	21.4	19.4	16.8	15.0	15.1	14.9	16.1	17.7	18.9	21.7	(22)
(23)		5%	DB	31.9	33.1	33.0	30.0	27.5	24.3	23.0	23.9	24.8	26.6	28.8	30.3	(23)
(24)			MCWB	23.5	23.9	22.4	20.5	17.6	15.8	15.2	15.5	16.9	18.3	19.9	22.3	(24)
(25)		10%	DB	30.3	31.5	31.2	28.5	25.8	23.0	21.8	22.5	23.4	25.1	26.9	28.7	(25)
(26)			MCWB	23.4	24.1	22.6	21.0	18.1	15.7	15.3	15.5	16.7	18.4	20.1	22.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.5	26.9	26.1	24.7	22.9	21.2	20.6	20.1	21.0	21.6	22.9	25.7	(27)
(28)			MCDWB	30.8	31.5	31.0	28.2	25.7	23.0	23.1	22.6	24.0	25.9	27.4	28.9	(28)
(29)		2%	WB	25.5	26.1	25.2	23.6	21.7	19.6	19.0	19.0	19.5	20.7	22.2	24.2	(29)
(30)			MCDWB	30.0	30.9	29.8	27.4	24.7	22.0	21.3	21.8	22.9	25.1	26.7	28.8	(30)
(31)		5%	WB	24.8	25.5	24.7	22.9	21.0	18.6	17.7	18.0	18.6	20.1	21.6	23.3	(31)
(32)			MCDWB	29.5	30.2	29.2	26.9	24.6	21.3	20.3	21.2	22.4	24.2	26.1	28.0	(32)
(33)		10%	WB	24.1	24.9	24.1	22.2	20.0	17.7	16.6	17.1	17.8	19.4	20.9	22.7	(33)
(34)			MCDWB	29.0	29.5	28.6	26.2	23.6	20.8	19.8	20.8	21.9	23.6	25.5	27.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	7.4	8.3	9.0	11.0	11.1	11.0	9.6	8.5	7.8	7.3	(35)	
(36)			MCDBR	10.8	11.1	12.2	12.4	14.0	13.5	13.2	14.1	12.6	12.3	12.2	10.3	(36)
(37)			MCWBR	4.5	4.5	5.5	6.0	7.1	6.9	7.0	7.6	6.6	6.4	5.7	4.6	(37)
(38)		5% WB	MCDBR	8.1	8.0	8.3	9.1	9.9	8.9	9.1	10.3	9.8	9.2	8.4	7.6	(38)
(39)			MCWBR	3.8	3.6	4.0	4.7	5.5	5.4	5.9	6.3	5.5	4.9	4.2	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.374	0.375	0.357	0.335	0.311	0.305	0.299	0.303	0.316	0.332	0.337	0.350	(40)	
(41)		taud	2.511	2.534	2.567	2.604	2.638	2.636	2.646	2.609	2.545	2.510	2.519	2.529	(41)	
(42)		Ebn at Noon	969	953	944	922	905	890	911	943	969	985	999	995	(42)	
(43)		Edh at Noon	114	109	101	89	79	75	77	87	101	111	113	113	(43)	
(44)	All-Sky Solar Radiation	RadAvg	8.16	7.57	6.55	5.31	4.33	3.75	3.99	5.04	6.36	7.41	8.17	8.51	(44)	
(45)		RadStd	0.33	0.29	0.38	0.26	0.26	0.17	0.28	0.19	0.15	0.24	0.23	0.33	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	-1.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	-1.75	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page